

Avanços da IA e Governança

Mapeando os Impactos Sociais e Éticos da IA via Knowledge Graphs

Bruno Paiva
Manoel Neto
Mariana Queiroz

Definição do Problema

- Fragmentação da informação em fontes multimodais (PDFs, vídeos e artigos)
- Como rastrear a influência real entre discursos de inovação e regras de governança?

Objetivo

- Utilizar o Graphify para identificar se a diversidade e inclusão impactam a produtividade e como resolver o dilema entre escala tecnológica e direitos éticos.



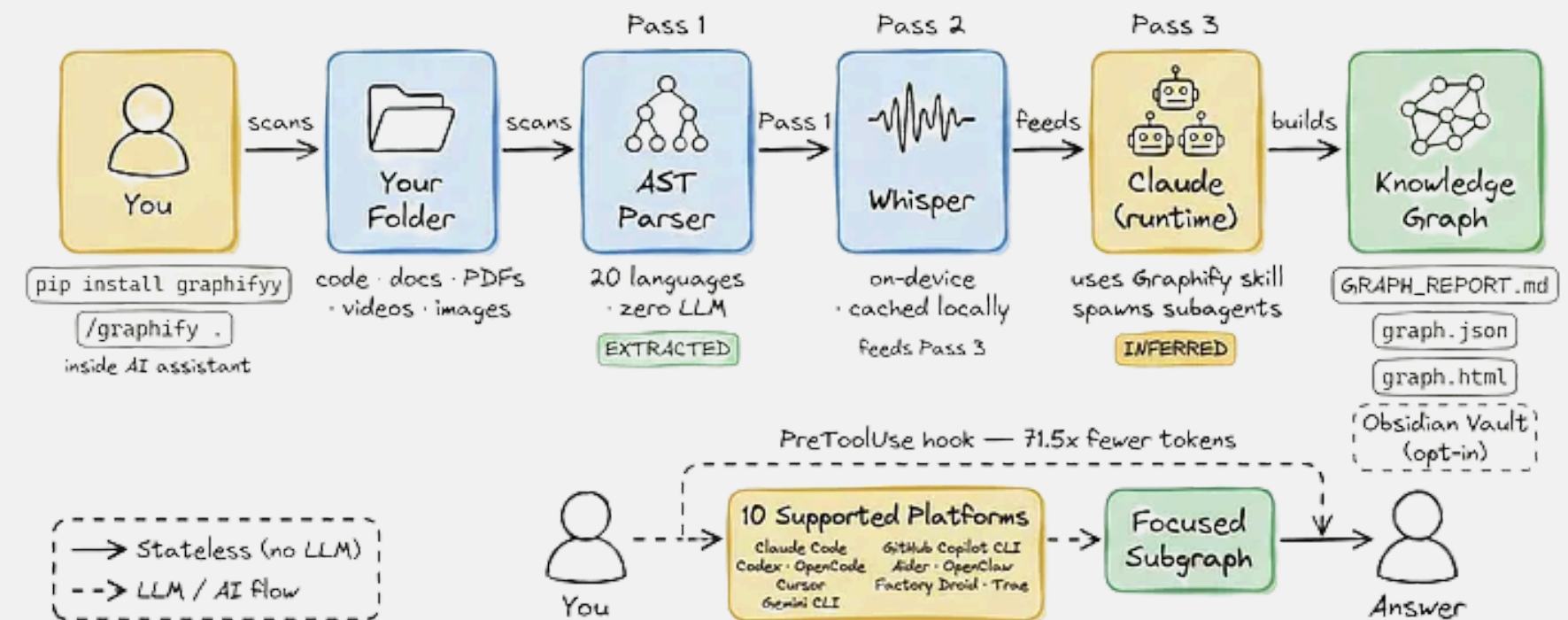
Abordagem

- Coleta de fontes heterogêneas (Relatórios da Goldman Sachs, vídeos sobre ética, artigos técnicos).
- Processamento via Graphify usando Codex.
- Processamento via NotebookLM

Metodologia de Comparação

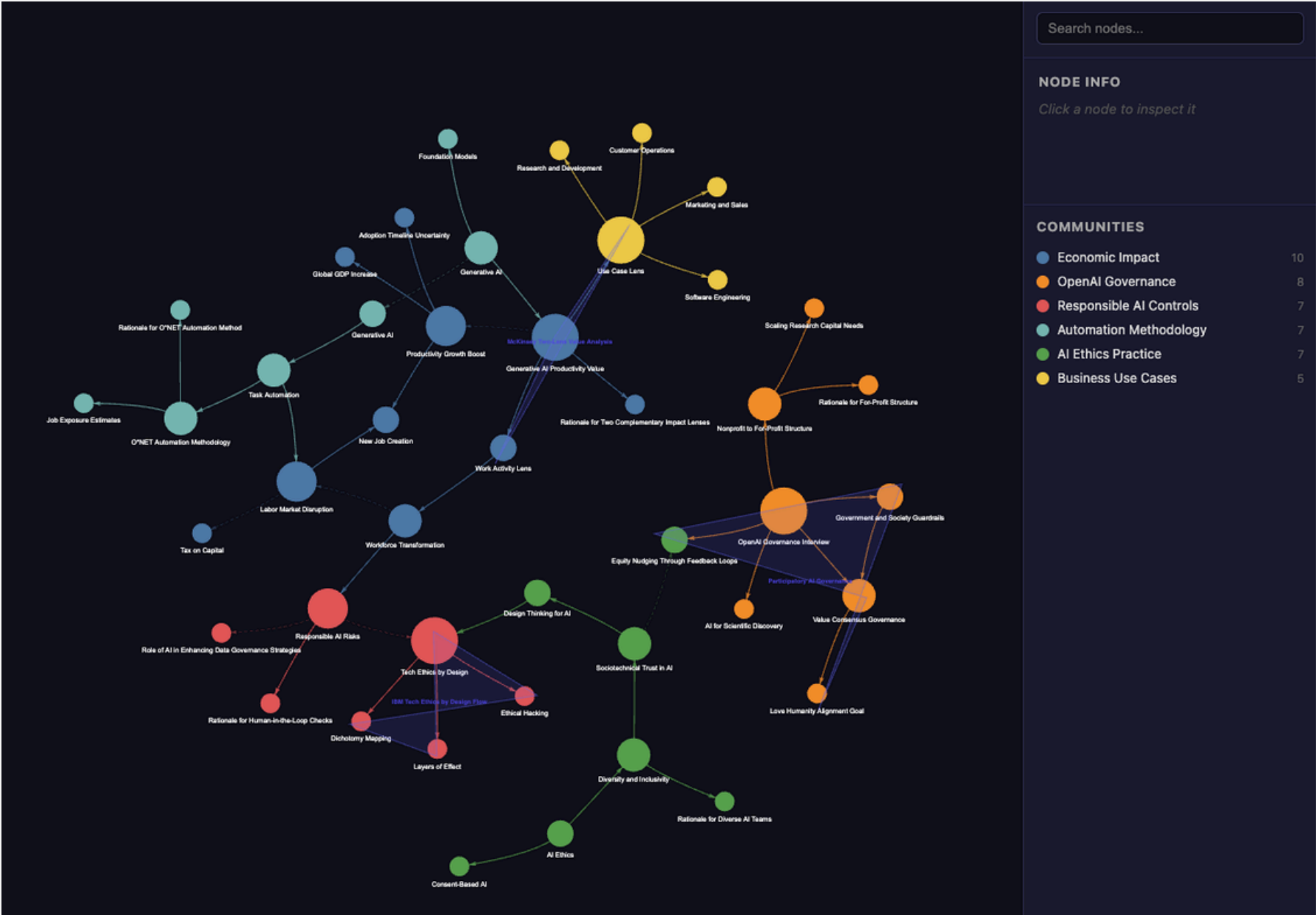
- Aplicação de prompts idênticos para observar como a estrutura do dado (Grafo vs. Texto Plano) altera a resposta final.
- Foi usado o Codex com GPT-5.5

Graphify: End-to-End Flow



Resultados

Graphify output



Graph Report - . (2026-04-28)

Corpus Check

- Corpus is ~13,739 words - fits in a single context window. You may not need a graph.

Summary

- 44 nodes · 46 edges · 6 communities detected
- Extraction: 85% EXTRACTED · 13% INFERRED · 2% AMBIGUOUS · INFERRED: 6 edges (avg confidence: 0.79)
- Token cost: 0 input · 0 output

Community Hubs (Navigation)

- [[_COMMUNITY_Economic Impact|Economic Impact]]
- [[_COMMUNITY_OpenAI Governance|OpenAI Governance]]
- [[_COMMUNITY_Responsible AI Controls|Responsible AI Controls]]
- [[_COMMUNITY_Automation Methodology|Automation Methodology]]
- [[_COMMUNITY_AI Ethics Practice|AI Ethics Practice]]
- [[_COMMUNITY_Business Use Cases|Business Use Cases]]

God Nodes (most connected - your core abstractions)

- Generative AI Productivity Value - 5 edges
- Use Case Lens - 5 edges
- Tech Ethics by Design - 5 edges
- OpenAI Governance Interview - 5 edges
- Responsible AI Risks - 4 edges
- Labor Market Disruption - 4 edges
- Productivity Growth Boost - 4 edges
- Generative AI - 3 edges
- Workforce Transformation - 3 edges
- Task Automation - 3 edges

Resultados

Prompts e respostas

According to the [GRAPH_REPORT.md](#), can diversity and inclusion add value and productivity to generative AI?

Yes, according to [GRAPH_REPORT.md](#), **diversity and inclusion can add value to generative AI indirectly**, by improving the ethical and social quality of AI development.

The report places **Diversity and Inclusivity** inside the **AI Ethics Practice** community, alongside **AI Ethics**, **Consent-Based AI**, **Design Thinking for AI**, **Sociotechnical Trust in AI**, and **Rationale for Diverse AI Teams**. This suggests that diverse teams help AI systems better reflect different user needs, social contexts, and possible harms.

However, the report does **not** show a direct edge from **Diversity and Inclusivity** to **Generative AI Productivity Value** or **Productivity Growth Boost**. So the strongest answer is: diversity and inclusion are not presented as a direct productivity mechanism in the graph, but as an ethical practice that can support better AI outcomes.

In practical terms, diversity and inclusion can add value to generative AI by helping teams design systems that are more trustworthy, less biased, more useful across different populations, and better aligned with social expectations. That trust and broader usability can make generative AI adoption more sustainable, which may support productivity over time.

Codex com Graphify

According to the sources, can diversity and inclusion add value and productivity to generative AI?

According to the sources, diversity and inclusion are not just ethical requirements but essential components that **add significant value and productivity** to the development and implementation of generative AI. This impact is seen through the reduction of technical errors, the improvement of project success rates, and the creation of more effective, human-centric systems.

1. Mathematical Reduction of Error

AI ethics is described as a "team sport" where exclusion breeds elitism and toxic notions ¹. From a technical standpoint, the sources highlight a **mathematical model** indicating that "the wider the variance, the more standard the mean" ¹. In other words, a **more diverse group of people** tackling a complicated problem results in a **lower chance for error** in the AI model ¹.

2. Overcoming the "Proof of Concept" Bottleneck

A major barrier to AI productivity is that approximately **80% of AI efforts get stuck in the "proof of concept" stage** ². This often occurs because the investments are not tied to business strategy or because people simply do not **trust the results** of the model ². Diverse teams help overcome this by:

- Using **design thinking** to tie AI models to clear business intent ³.
- Cracking the **"empathy code"** before any programming code is written, which ensures the technology is better aligned with user needs ³.
- Planning the rollout of AI efforts through a culture that values responsible, consent-based data usage ⁴ ⁵.

3. Enabling "Systemic Empathy" and Human Augmentation

Adopting frameworks for **systemic empathy** allows organizations to create AI that is **intentional in its efforts to augment human beings** rather than just replacing them ⁶ ⁷. Truly diverse teams—incorporating different **genders, races, ethnicities, ages, neurodiversities, worldviews, and skill sets**—are required to achieve this systemic equity ¹ ³. By understanding what a user is "thinking, seeing, hearing and doing," these teams can identify potential **tertiary effects** (unintended harms) and mitigate them during the design phase ⁶ ⁷.

4. Contributing to Inclusive Economic Growth

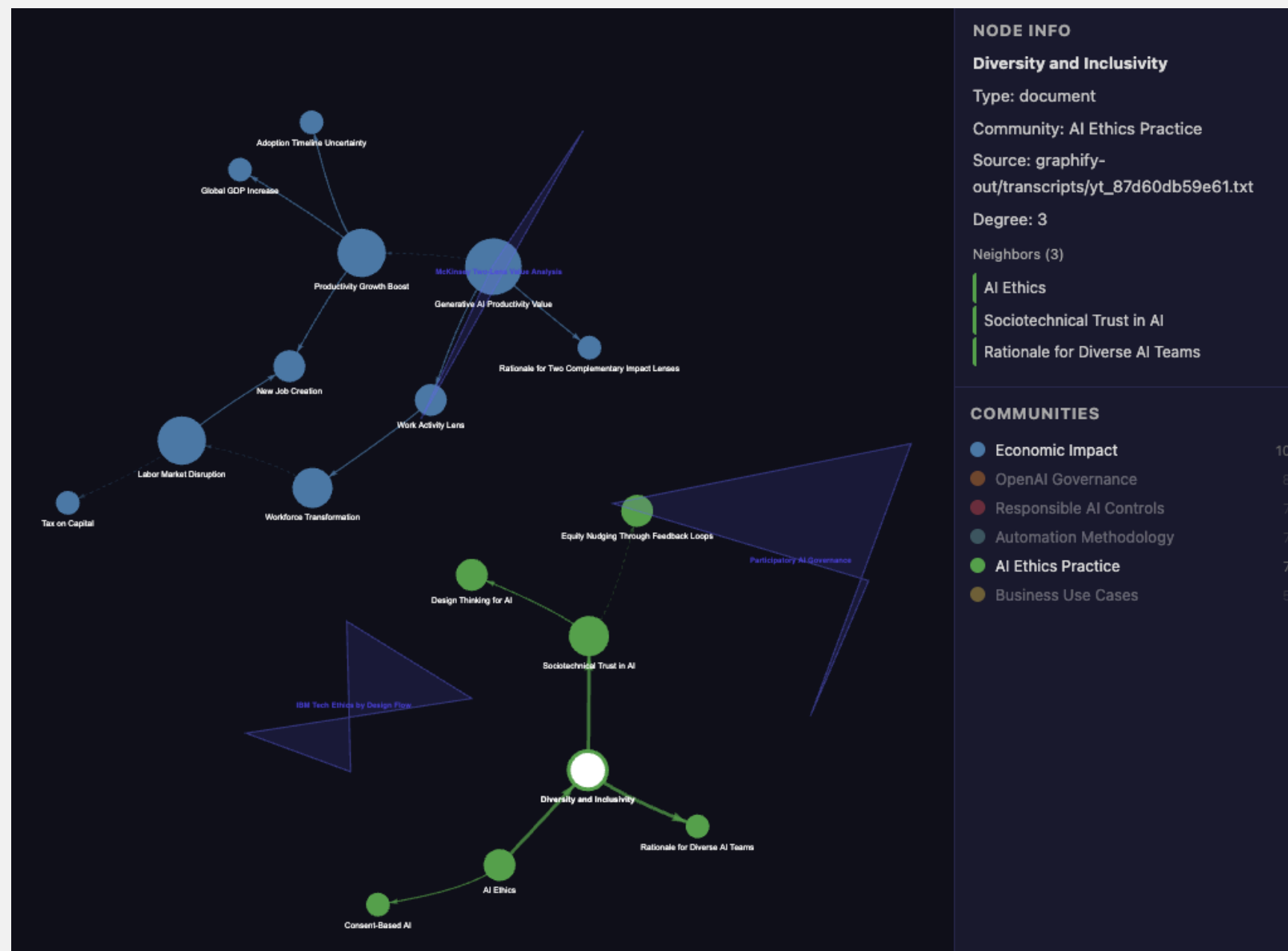
The integration of diverse perspectives ensures that AI development accounts for **societal values**, which is critical for the long-term success of the technology ⁹. If worker transitions and ethical risks are managed through these inclusive practices, generative AI can contribute substantively to economic growth and support a more **sustainable and inclusive world** ¹⁰.

In summary, diversity provides the **Ethos** (ethical standards) necessary to democratize AI, ensuring it serves as a "virtual expert" or "collaborator" that functions openly and justly across the global economy ⁷ ^{...}.

Resultados

Prompts e respostas

According to the sources, can diversity and inclusion add value and productivity to generative AI?



- O Graphify identificou:
 - Não existe uma "aresta" direta entre Diversidade e Aumento de Produtividade.
 - Há uma relação indireta.
- No mapa do conhecimento, a diversidade não "gera" produtividade sozinha; ela "gera" Confiança (Trust), que por sua vez é o que permite que a Produtividade seja destravada.
- O NotebookLM extraiu a justificativa técnica: "quanto maior a variância (diversidade), menor a chance de erro (média padrão)". Isso transforma diversidade em uma métrica de QA (Quality Assurance).
- Ele aponta que 80% dos projetos de IA falham na fase de prova de conceito por falta de confiança. A diversidade resolve o "código de empatia", permitindo que o projeto avance para a produção.
- A diversidade aumenta a produtividade ao reduzir o retrabalho e o abandono de projetos.

Resultados

Prompts e respostas

based on [GRAPH_REPORT.md](#) What is the dilemma between the advancement of AI and guaranteeing the ethical rights of users and society?

Based on [GRAPH_REPORT.md](#), the dilemma is that AI advancement promises major economic and productivity gains, but those gains depend on systems that can also disrupt labor, concentrate power, and affect people's rights at scale.

On one side, the graph highlights the advance of AI through **Generative AI, Foundation Models, Task Automation, Productivity Growth Boost, Generative AI Productivity Value, Business Use Cases, and Global GDP Increase**. These nodes show AI as a driver of automation, efficiency, new products, and economic growth.

On the other side, the graph shows ethical and social concerns through **Responsible AI Risks, Human-in-the-Loop Checks, Tech Ethics by Design, AI Ethics, Consent-Based AI, Diversity and Inclusivity, Sociotechnical Trust in AI, and Government and Society Guardrails**. These nodes point to the need to protect users and society from harm, bias, exclusion, loss of consent, weak accountability, and untrustworthy deployment.

The core dilemma is this: **the faster AI systems are scaled for productivity and economic value, the harder it becomes to ensure that they remain fair, transparent, consent-based, inclusive, and socially accountable**. If governance is too weak, AI can amplify inequality, automate decisions without adequate human oversight, and transform work without protecting affected people. But if governance is treated only as restriction, organizations may see it as slowing innovation.

The graph suggests a middle path: ethical rights should not be added after AI is deployed. They need to be built into the design and governance of AI systems through **Responsible AI Controls** and **AI Ethics Practice**. In that sense, the dilemma is not simply "innovation versus ethics," but whether society can make AI advancement depend on ethical design, human oversight, diverse participation, and public guardrails from the beginning.

According to the sources, what is the relationship between the risk of responsible AI and the economic impact of AI?

According to the sources, the relationship between the risk of responsible AI and its economic impact is one of **interdependence**: the massive economic potential of AI can only be fully realized if the ethical and societal risks are proactively managed through responsible governance.

The sources highlight this relationship through several key points:

1. Trust as a Prerequisite for Economic Value

A significant barrier to AI's economic impact is the lack of trust. The sources indicate that approximately **80% of AI efforts fail to move past the "proof of concept" stage** because they are either not tied to business strategy or because users and stakeholders simply do not trust the results ¹. For AI to drive its projected **\$7 trillion increase in annual global GDP**, organizations must first overcome the "sociotechnological challenge" of earning trust through responsible design and systemic empathy ² ³.

2. The Innovation-Regulation Balance

There is a direct tension between the speed of economic innovation and the implementation of ethical guardrails.

- **The Risk of Over-regulation:** Excessive regulation can **slow down innovation**, potentially delaying the labor productivity boom that AI is expected to catalyze ⁴.
- **The Risk of Under-regulation:** Conversely, insufficient regulation can lead to "a whole set of other problems," including data breaches, discriminatory outcomes, and a "hollowing out" of middle-income job sectors, which would undermine long-term economic stability ⁴.

3. Managing Labor Disruption to Capture Growth

While AI could raise annual labor productivity growth by **1.5 percentage points** over a decade, it also exposes the equivalent of **300 million full-time jobs** to automation ⁷ ⁸. The sources state that generative AI will contribute substantively to economic growth **only "if worker transitions and other risks can be managed"** ⁹. Failure to handle this transition responsibly—such as neglecting reskilling programs—poses a risk to the very sustainability and inclusivity of that economic growth ⁹ ¹⁰.

4. Data Governance as Financial Protection

From a legal and operational standpoint, responsible AI practices like **data governance** are essential to protect the economic bottom line. Adhering to frameworks like **GDPR and CCPA** helps mitigate risks related to:

- **Legal Sanctions:** Non-compliance can lead to massive fines (up to €20 million or 4% of worldwide annual sales under GDPR) ⁶ ...
- **Reputational Harm:** Unauthorized access or data breaches can cause lasting damage to consumer trust, which is critical for the "safe and equitable operation" of AI-driven business ⁶ ¹³.

5. Technical Performance through Diversity

The sources argue that **diversity and inclusion** act as a technical safeguard that enhances productivity. By employing diverse teams, organizations can use a "mathematical model" to reduce the chance of errors and bias in AI models ¹⁴. This ensures the technology is more accurate and aligned with user needs, thereby making the AI's output more valuable and effective in a business context ¹⁵.

In summary, the sources suggest that **Responsible AI is not a hurdle to economic impact, but a foundation for it**. Organizations that prioritize "tech ethics by design" are better positioned to leverage AI's full potential while minimizing the security, compliance, and ethical issues that could otherwise derail their investments ¹⁶ ¹⁷.

Resultados

What is the dilemma between the advancement of AI and guaranteeing the ethical rights of users and society?

Prompts e respostas



- O Graphify identificou uma tensão entre polos opostos:
 - Polo de Avanço (Cluster A): Focado em Generative AI, GDP Increase e Task Automation. Representa o vetor de aceleração econômica.
 - Polo Ético (Cluster B): Focado em Consent-Based AI, Guardrails e AI Ethics. Representa o vetor de proteção social.
- O nó de "Responsible AI Risk" atua como uma ponte obrigatória. Sem passar por ele, a inovação gera riscos que isolam a tecnologia da sociedade.
- No NotebookLM, o LLM fornece a profundidade narrativa das tensões identificadas no grafo:
 - Explica o conflito técnico de que modelos mais potentes (parâmetros em bilhões) tendem a ser menos explicáveis.
 - Aponta que a necessidade de alto capital empurra a IA para o setor privado, criando um dilema de governança sobre quem controla a segurança pública.
 - Detalha que regulação excessiva pode travar o "boom de produtividade", enquanto a falta dela gera "hollowing out" (esvaziamento) de setores de renda média.

Limitações

- Para encontrar relações visualmente mais fortes, é interessante selecionar fontes com contextos semelhantes.
- O Grafo mostra que temas estão ligados, mas não diz com precisão "quem influenciou quem" cronologicamente.
- Fontes de vídeo exigem um pré-processamento de limpeza mais rigoroso para não gerar nós equivocados no Knowledge Graph.
- O grafo pode simplificar demais relações que o LLM descreve com mais nuance e tom crítico.

Obrigado.

Código fonte, referências e outputs: <https://github.com/brunopaiva1/ai-governance>

Bruno Paiva
Manoel Neto
Mariana Queiroz